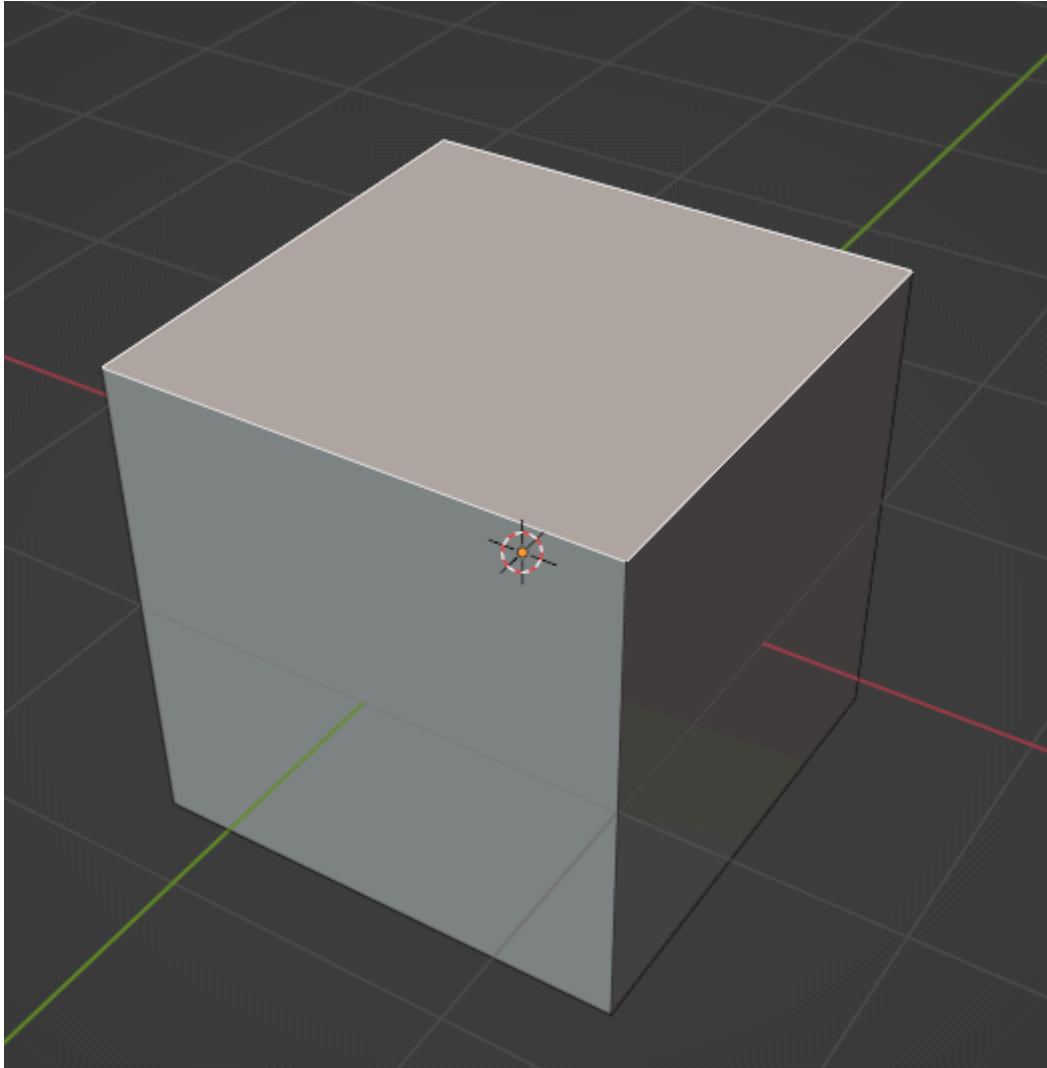


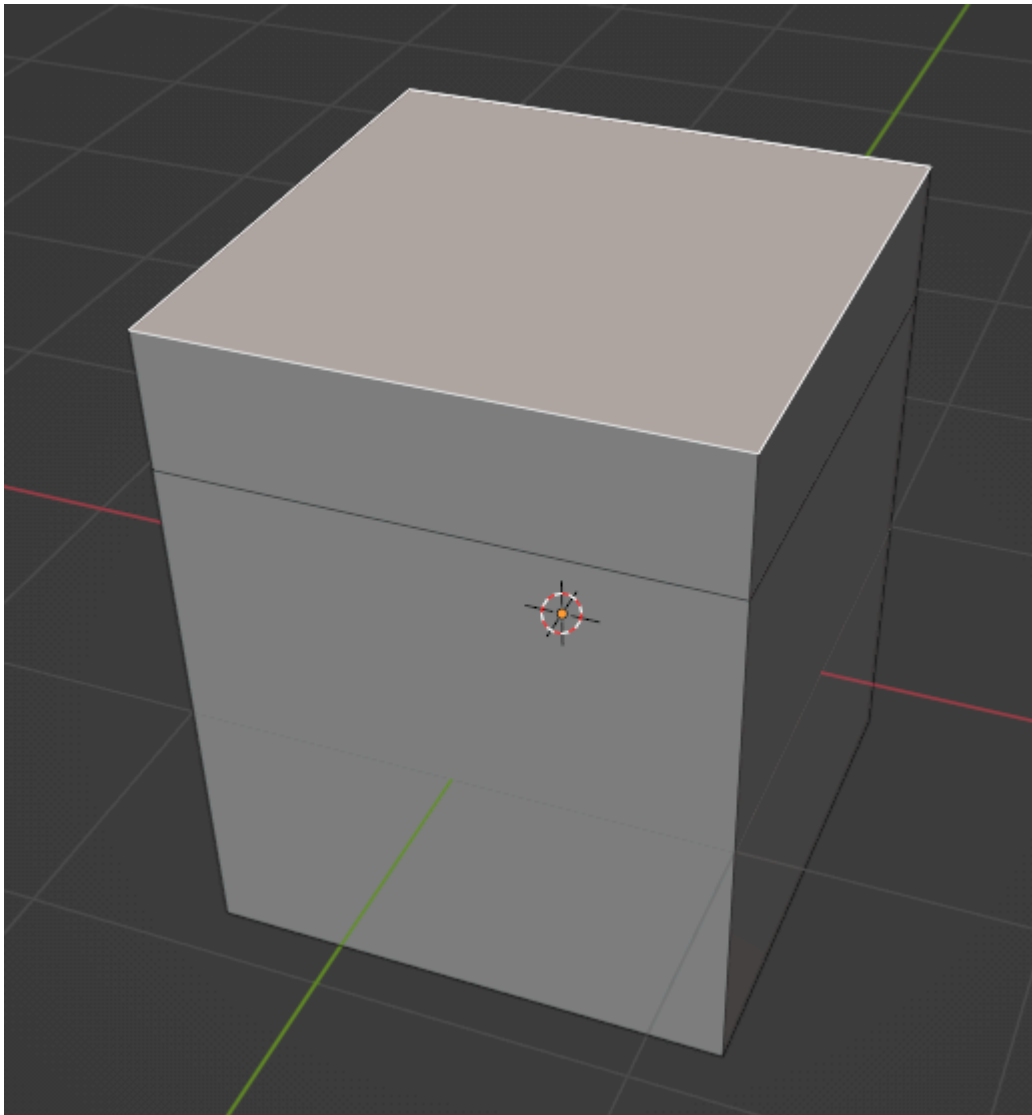
Extrude Tutorial

Important concepts covered: extrude

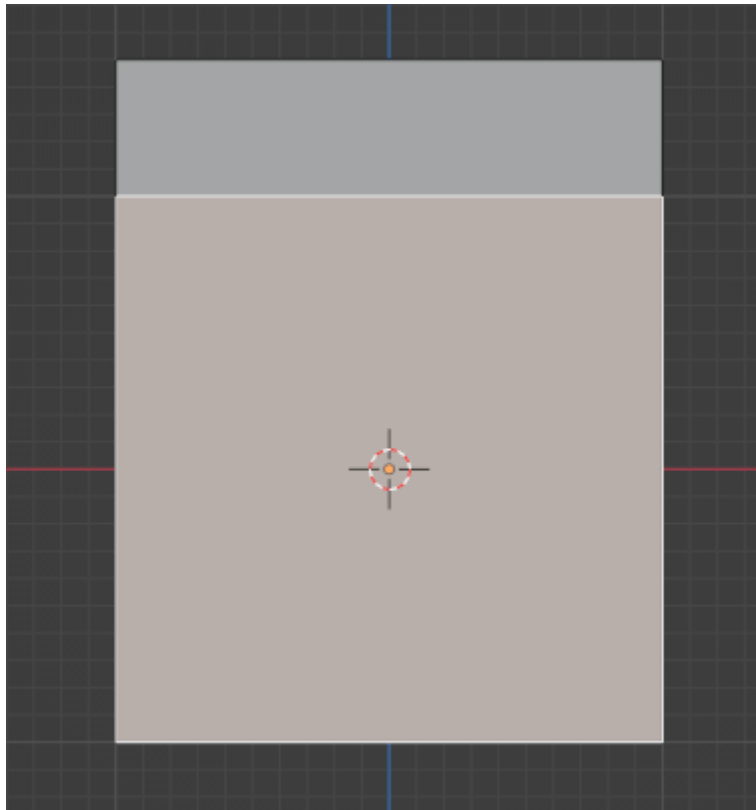
Start by entering edit mode in the default cube. Then change to face selection mode and select the top face:



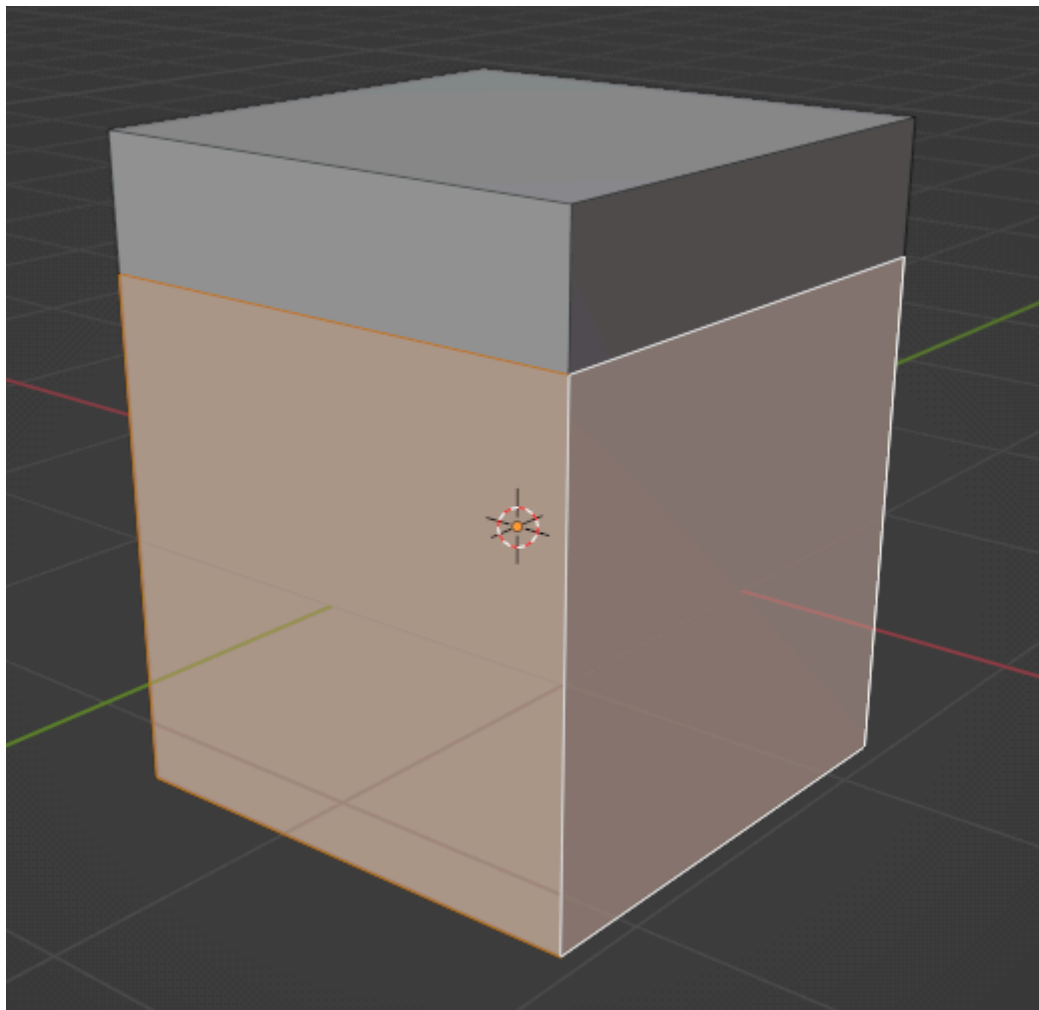
To extrude, you can press “E” and type 0.5:



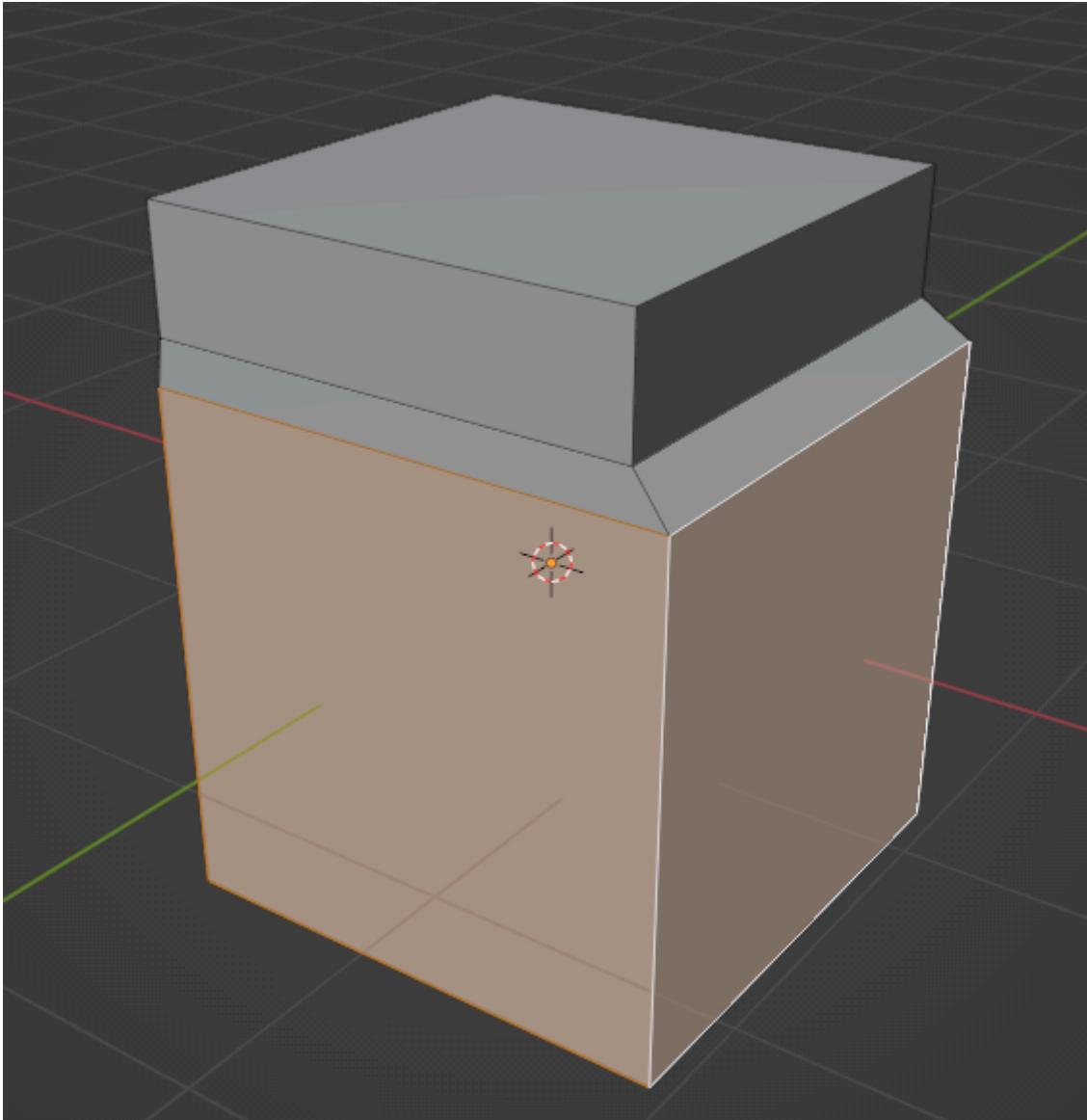
This is the standard extrude operation. It will automatically extrude along the normals of the face. If you select 2 faces, it will calculate the combined normal vectors of both selections and move along this direction. For example: press NUMPAD 1 to go to the front view and select the front bottom face:



Now select also the face connected to this face on its right side:

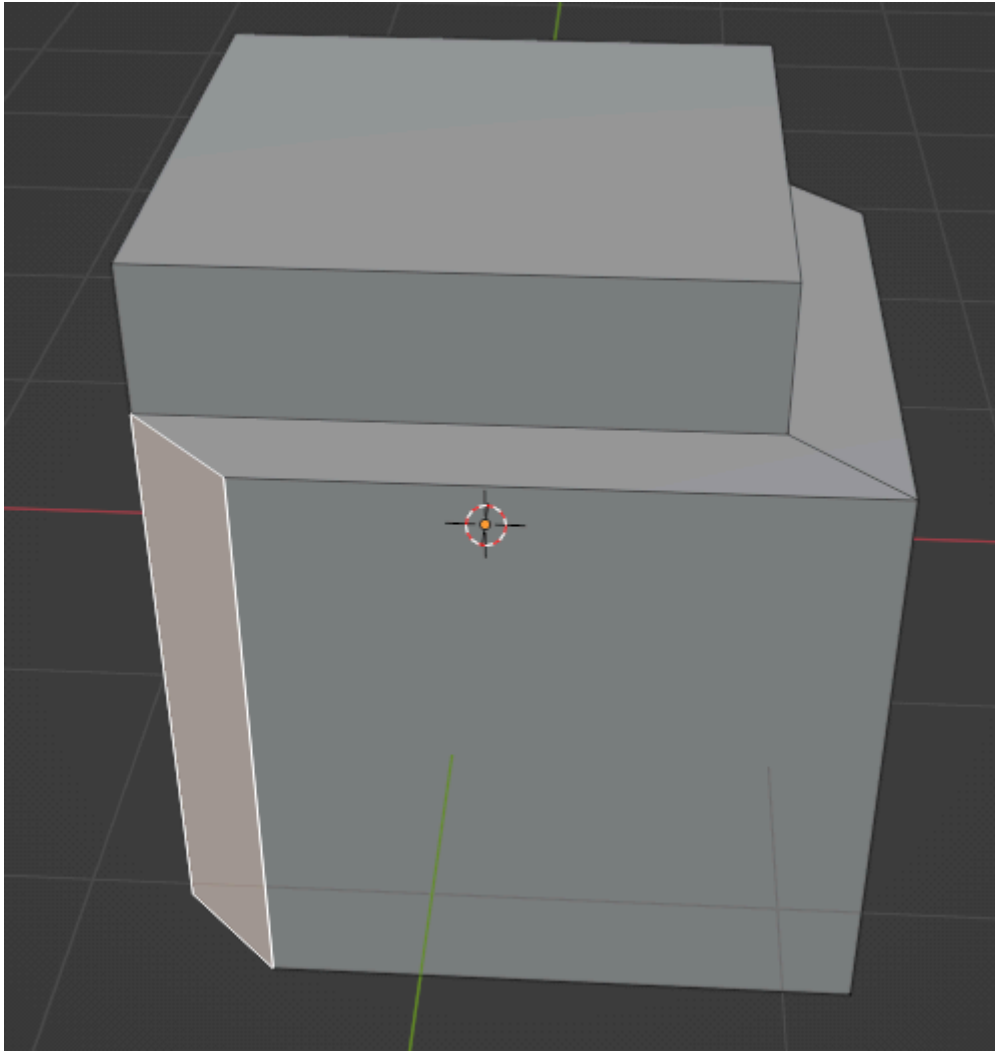


Now press “E” again to extrude and type 0.5 again:

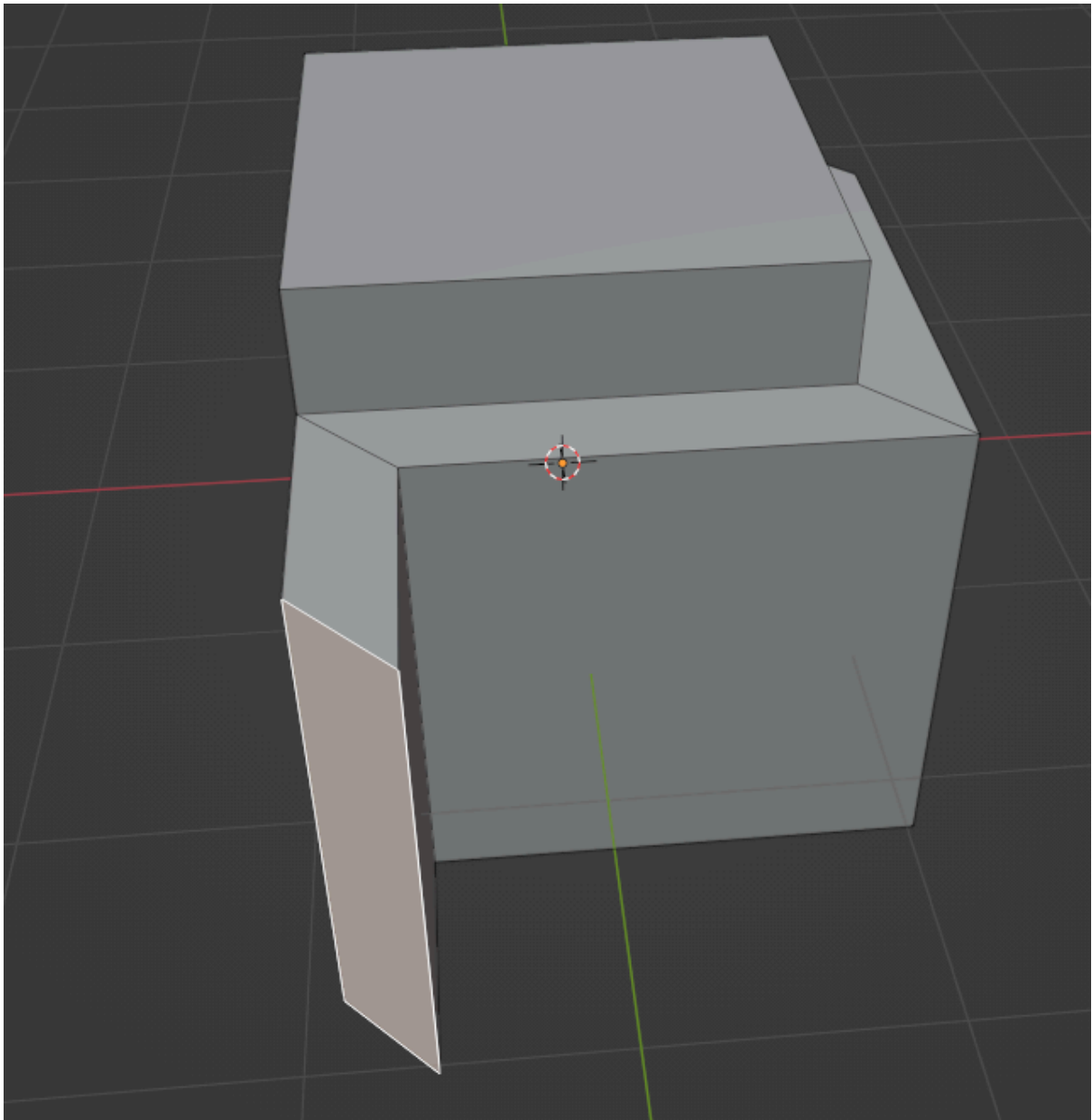


As you can see it will follow a diagonal direction, which is the combination of the 2 normals of the faces.

Now, let's specify the direction. Select the following face:

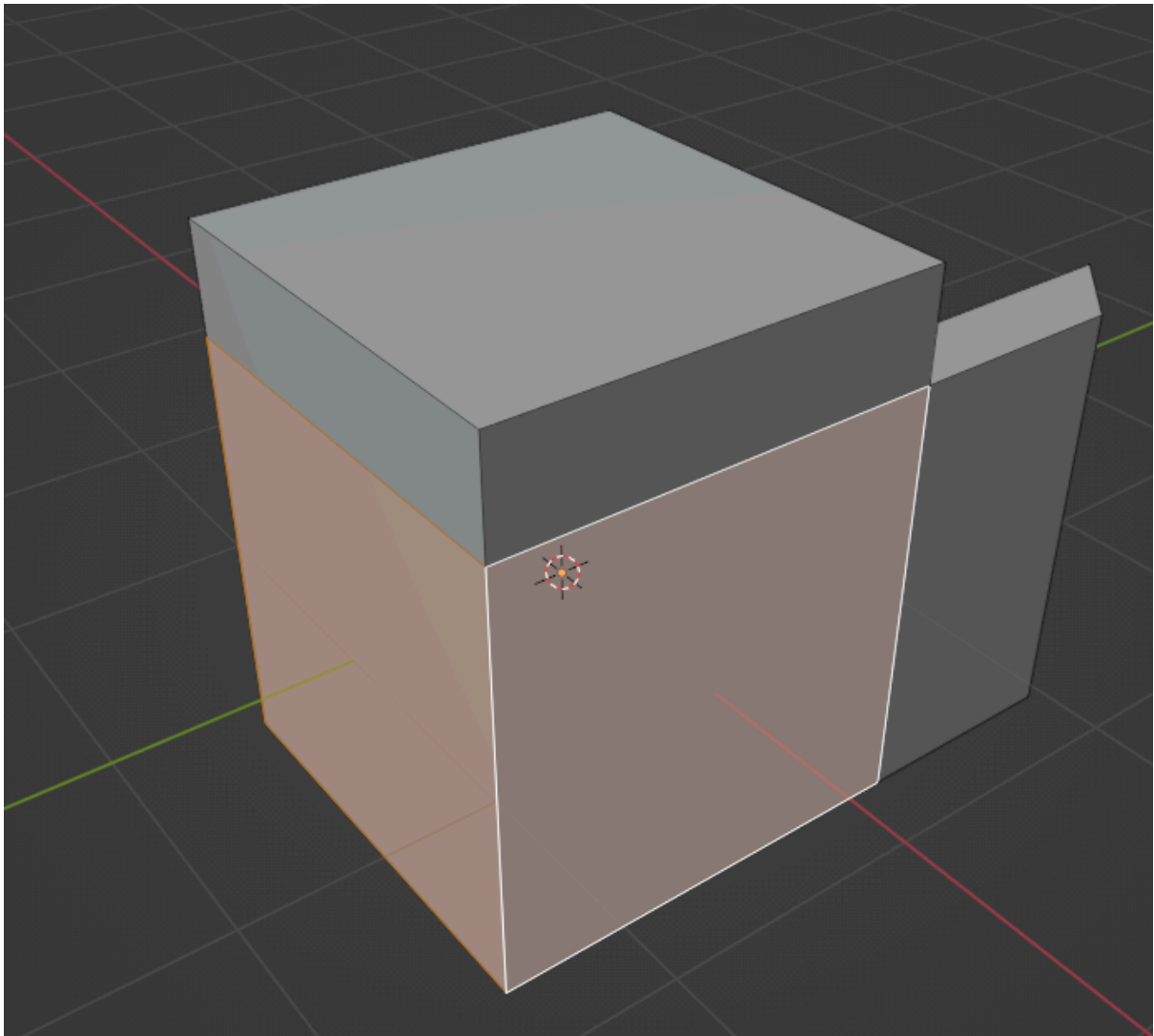


Extrude by pressing “E”, then press “Y” and then type -1:

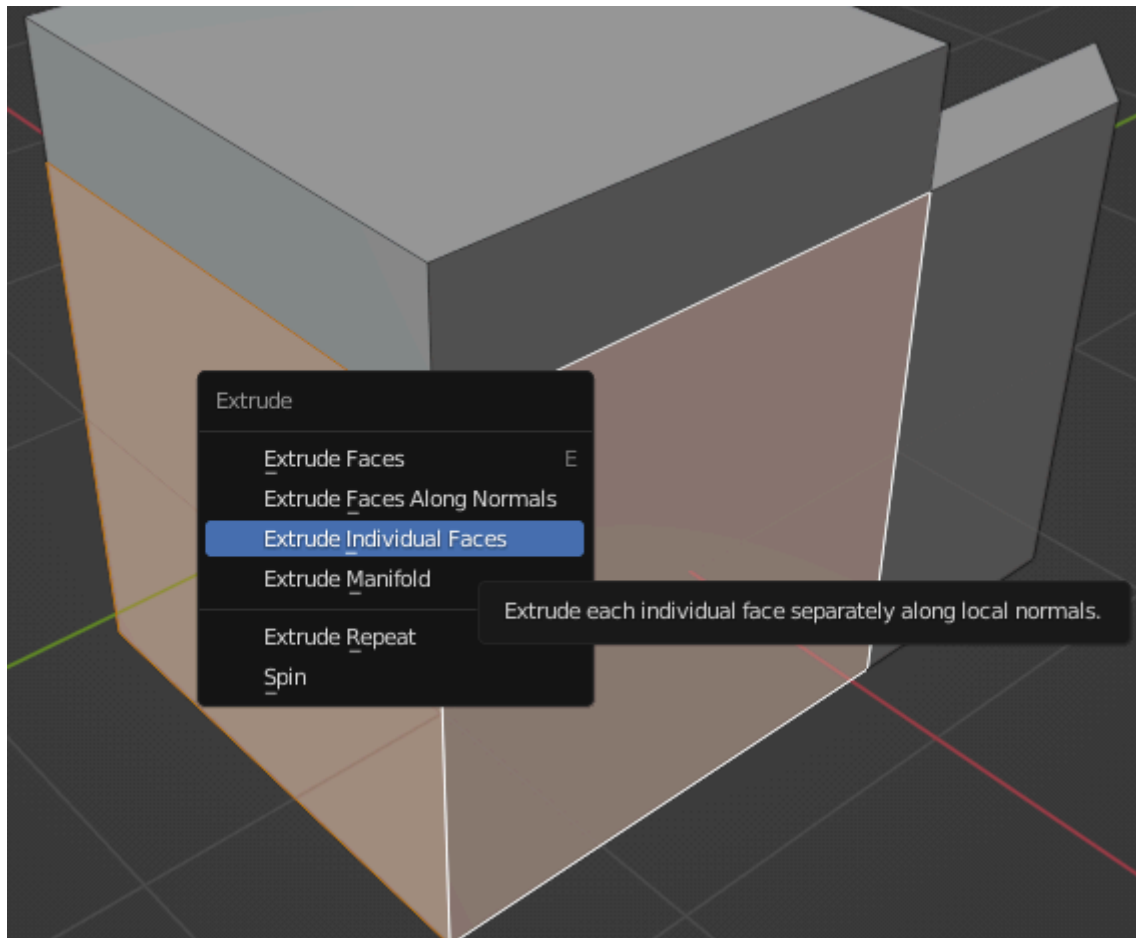


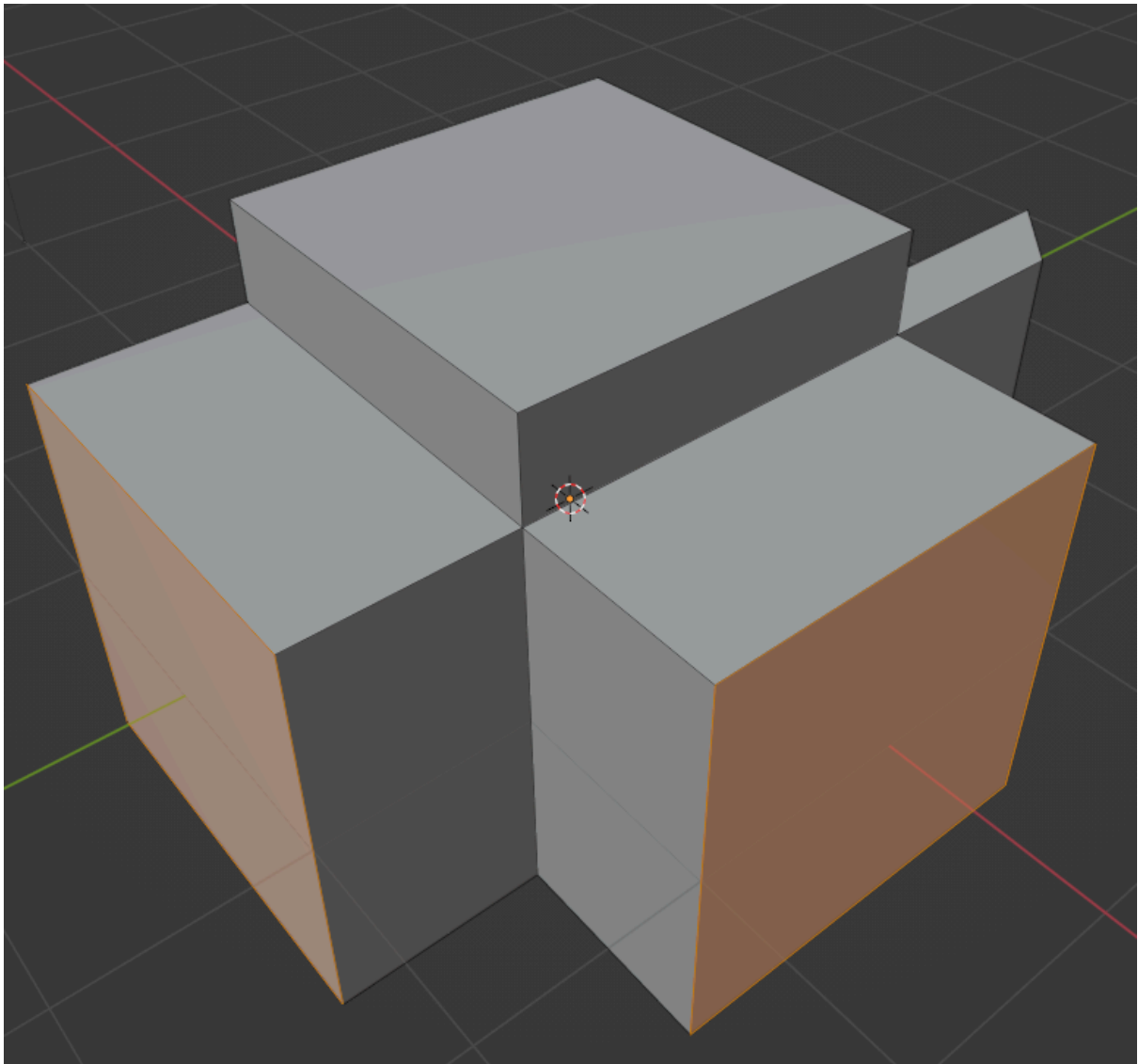
As you can see, now it has been extruded following the Y axis. You can do it for all axes by pressing “X”, “Y” or “Z”.

But what if you want to extrude individual faces with their individual normals? To do this, select the other 2 faces from the back:



Now, press “ALT+E” and choose “Extrude Individual Faces” and type 1:

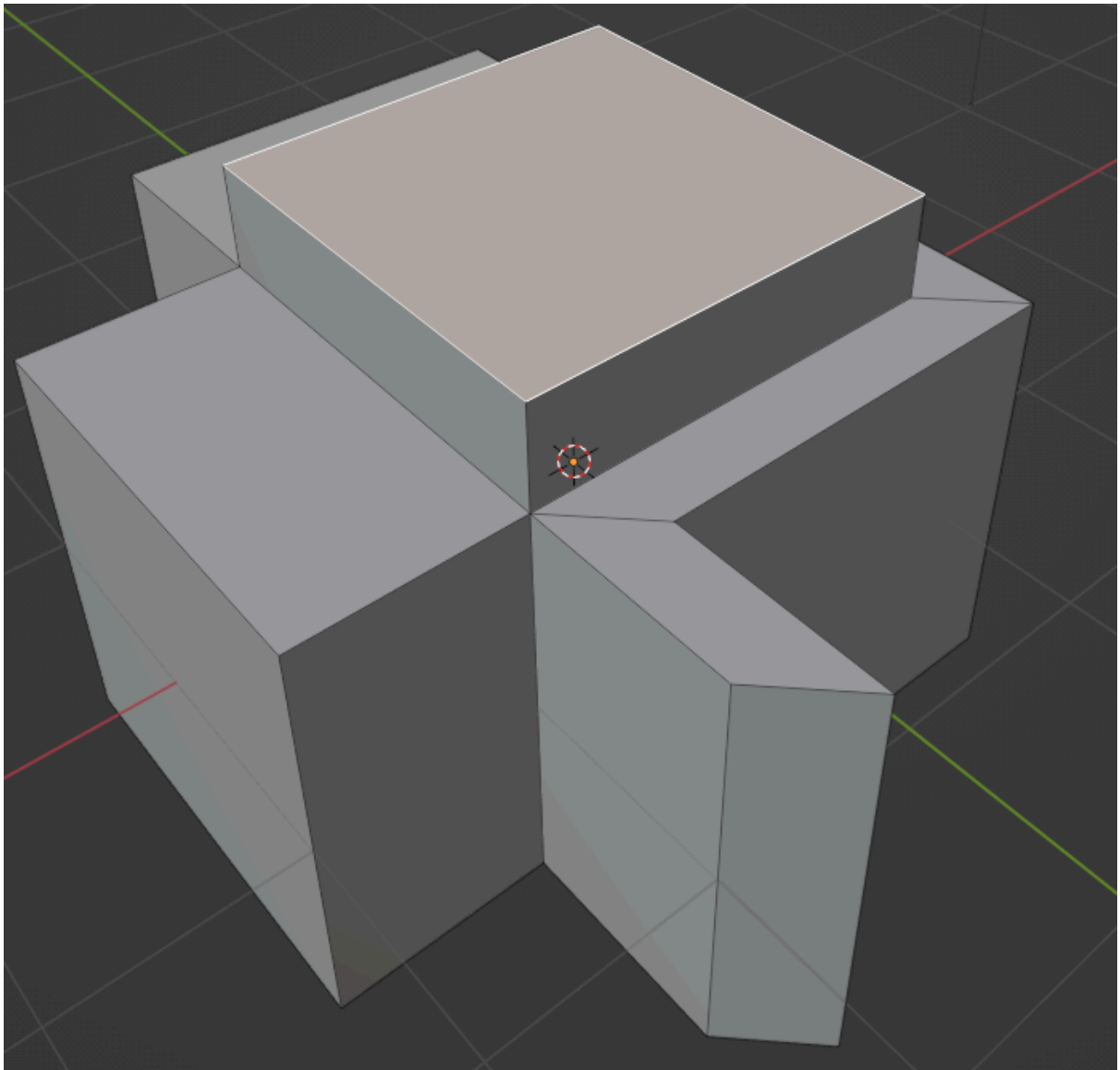




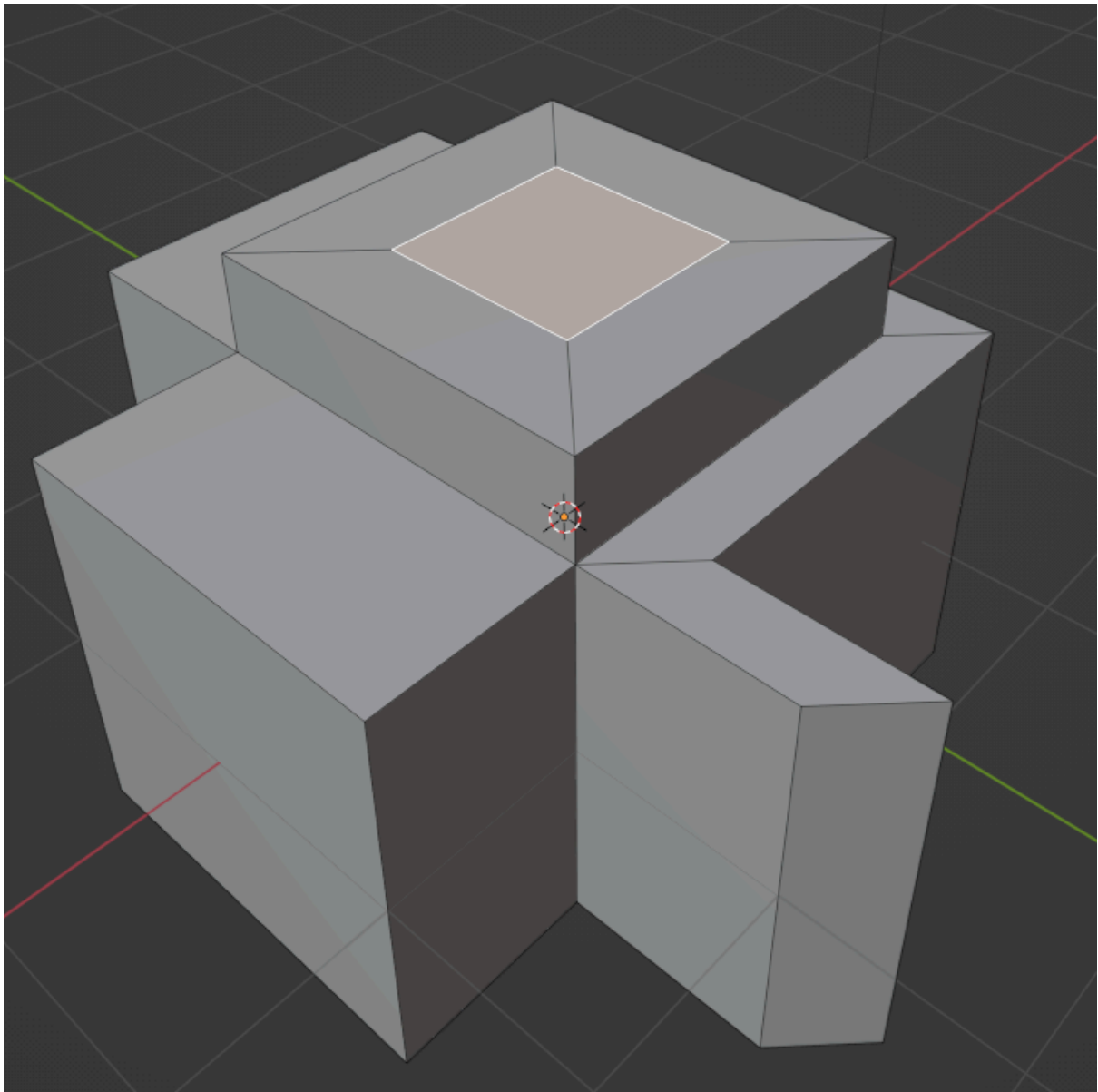
As you can see, the faces were extruded individually. You can also specify the axis in this case in the same way as before.

Finally, another useful thing to know is how to extrude and scale instead of moving the face along the normals.

To do this, select the top face:



Press “E” and then press “S” and then type 0.5:



This will tell blender that you want to scale the extruded face instead of move it (which is the default). You can also tell blender to rotate instead of move or scale, by pressing “R”. Then, if you type something, it will be translated to degrees by blender.

This concludes the extrude tutorial.